



# 7CS030 – Concepts & Technologies of Artificial Intelligence

School of Mathematics and Computer Science

Faculty of Science and Engineering

Coursework Assignment

on

Machine Learning Applications in Real-World Data Analysis:  
Regression, Clustering, Classification, and Ethical  
Considerations in AI

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## Task 1. House Price Prediction Report using Linear Regression

### 1.1 Introduction

This report details our model for predicting house prices using features like house size, material quality, and neighborhood characteristics. Our aim is to help you make informed pricing decisions by understanding which features most impact prices and how.

### 1.2 Understanding the Data

Our dataset includes 21,613 properties with 18 features (e.g., bedrooms, bathrooms, and living area size). We began by examining how each feature correlated with price to determine which would best inform our model. The most relevant features were:

- **House size (sqft\_living):** Larger homes tend to sell for more.
- **Quality (grade):** Homes with higher-quality materials and finishes are typically valued higher.

### 1.3 Building the Prediction Model

The codes for the models discussed here are available in the accompanying **Task 1** python file.

#### 1.3.1 Starting with One Feature

We first built a basic model using only house size (sqft\_living) as a predictor, utilizing Linear Regression.

- **How We Checked Accuracy:** The model's R-squared ( $R^2$ ), also called the coefficient of determination was 0.50, meaning it explained 50% of the price variations. While useful, there's room for further improvement.

#### 1.3.2 Expanding to Multiple Features

Next, we added quality (grade) to create a two-feature model. This increased the  $R^2$  to 0.55, explaining 55% of price variations and slightly improving accuracy. To assess the model's reliability further, we ran the two-feature model **100** times, with each iteration using a **different random state**. Averaging the R-squared values across these iterations gave us an  $R^2$  of 0.53, suggesting consistent performance and reliability.

### 1.4 Model Comparison

- **Single Feature:** The one-feature model is straightforward, predicting price based on size alone.
- **Multiple Features:** Including quality offers a richer perspective on how size and quality influence price, though the improvement is modest.

Depending on your needs, either model could work: the one-feature model is simpler, while the multi-feature model provides added context.

### 1.5 Visualizing the Findings

To illustrate how features affect price, we created a 3D plot as seen in **Figure 1** below. Each dot represents a house, with its vertical position showing price, and the two horizontal axes representing house size and quality. The red plane indicates our predicted prices. Houses near this plane match our predictions, while those above or below it differs, showing how price can vary with size and quality.

In short, larger and higher-quality homes tend to have higher prices, revealing the influence of these features on pricing.

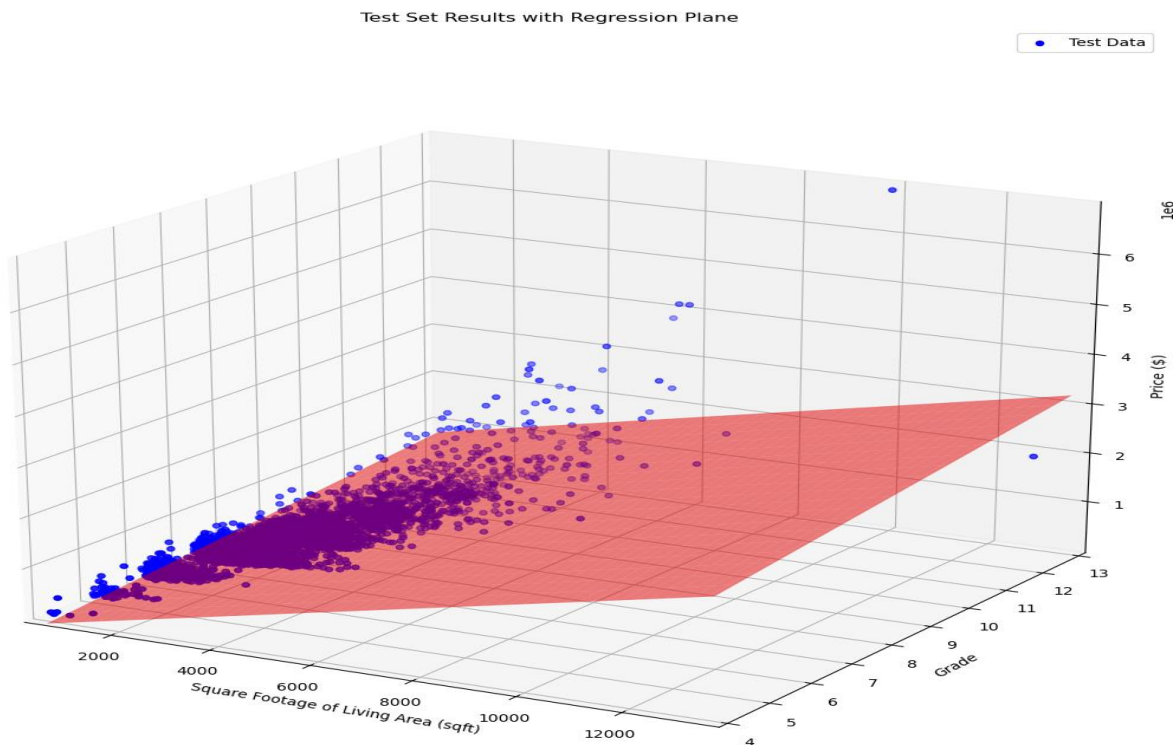


Figure 1. 3D Plot of Regression Plane for House Prices by Living Area and Grade

### 1.6 Key Takeaways and Recommendations

1. **Prioritize House Size and Quality:** These factors significantly drive price, so consider both when making pricing decisions.
2. **Model Choice Based on Needs:**
  - The **one-feature model** is simpler.
  - The **multi-feature model** offers a broader perspective with slightly added complexity.

### 1.7 Conclusion

We examined two models: a single-feature model and a multi-feature model. Both offer valuable insights, with the multi-feature model providing a broader view of how size and quality influence pricing. This model serves as a useful guide for price estimation, highlighting key market drivers. The consistency observed in the multi-feature model, as evidenced by the averaged R-squared score of 0.53, demonstrates its reliability.

### 1.8 Further Enhancements to Consider

So far, we explored how features like home size (sqft\_living) and quality (grade) affect price when looked at together. However, some relationships may be more complex. For example, as a home's size increases, the impact on price might grow faster for very large homes, not just in a simple straight line.

To capture this, we could introduce something called polynomial features. Think of it as adding features like "size squared"—a way to give extra emphasis to larger homes. This would allow the model to pick up on patterns where price rises more sharply for very large or high-quality homes. By considering these more subtle patterns, the model could better match real-world prices, potentially creating more accurate models.

## Task 2: Country Clustering Analysis Report

### 2.1 Introduction

This report analyzes countries by grouping them based on socio-economic and health indicators. By clustering countries with similar characteristics, we can identify patterns to better target economic and healthcare strategies suited to each country's needs.

### 2.2 Understanding the Data

Our dataset includes indicators like income, child mortality, life expectancy, health spending, and GDP per capita. Each of these features provides a window into a country's financial and healthcare status, helping us group similar countries.

### 2.3 Why Clustering?

Clustering groups countries by shared characteristics, uncovering hidden socio-economic patterns and enabling effective resource targeting where it's needed most.

### The Clustering Model: KMeans

Using KMeans, each country is assigned to a cluster, grouping similar profiles to provide meaningful insights.

The codes for the models discussed here are available in the accompanying **Task 2** python file.

### 2.4 Analysis 1: Income and Child Mortality Clustering

To understand socio-economic disparities, we started by clustering countries based on **income** (reflects financial stability, affecting healthcare, housing, and nutrition.) and **child mortality** (shows healthcare quality)

### Expected Clusters

We anticipated four clusters:

1. **Low Income, Very High Child Mortality:** Countries with limited resources.
2. **Low to Middle Income, Moderate to High Child Mortality:** Transitional economies with healthcare challenges.
3. **Upper Middle Income, Moderate Child Mortality:** Emerging economies with better healthcare.
4. **High Income, Low Child Mortality:** Wealthier countries with strong healthcare systems.

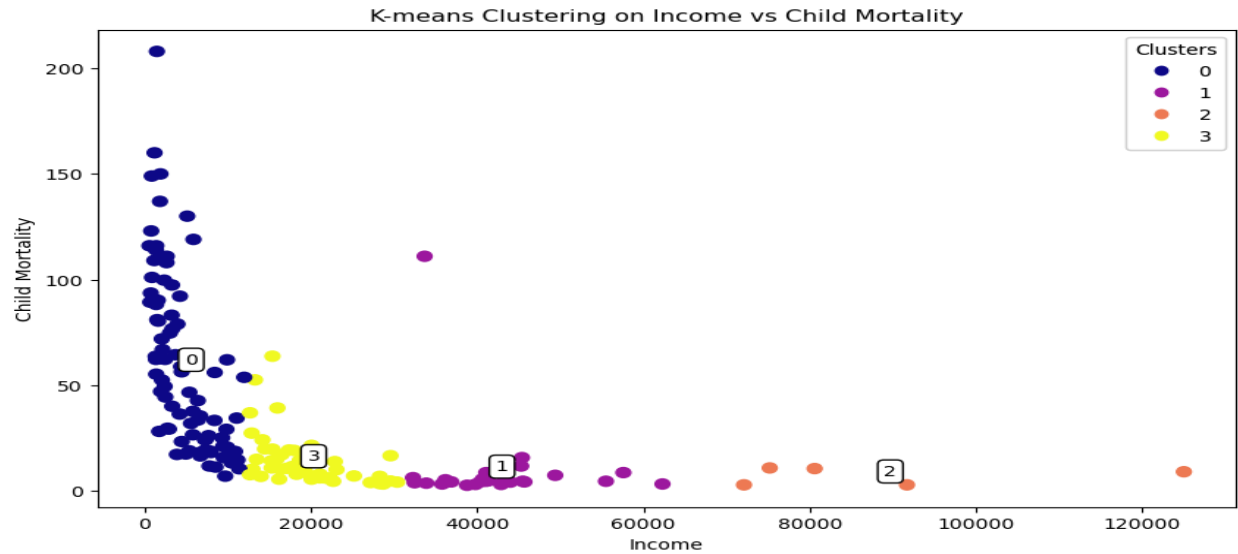
### Clustering Results

Our analysis identified four groups:

1. **Cluster 0 (Blue):** Low-income countries with very high child mortality, needing healthcare support.
2. **Cluster 1 (Purple):** Middle-income countries with moderate child mortality, often emerging economies.
3. **Cluster 2 (Orange):** High-income, low-mortality countries with advanced healthcare.
4. **Cluster 3 (Yellow):** Low to middle-income countries facing healthcare challenges.

### Visualization

Figure 2 below is a 2D chart with income and child mortality on the axes. Each cluster is represented by a color, making it easy to see which countries are grouped by their economic and healthcare status.



**Figure 2:** K-means Clustering of Income and Child Mortality

#### Recommendations

- **Cluster 0:** Needs international aid to improve healthcare access.
- **Clusters 1 and 3:** Investment in healthcare and economic growth can aid stability.
- **Cluster 2:** Continuing current policies supports sustained health standards.

### 2.5 Analysis 2: Life Expectancy, Health Expenditure, and GDP Clustering

For a broader view, we analyzed countries based on **life expectancy**, **health spending** (showing investment in healthcare), and **GDP per capita** (indicating financial capability to support services.)

**Expected Clusters:** We anticipated three clusters

1. Low Life Expectancy, Low Health Spending, Low GDP: Countries with limited resources.
2. Moderate Life Expectancy, Moderate Health Spending, Moderate GDP: Economies improving healthcare and quality of life.
3. High Life Expectancy, High Health Spending, High GDP: Wealthier countries with strong healthcare systems.

#### Clustering Results

The model identified three clusters:

1. **Cluster 0:** Low life expectancy and low health spending.
2. **Cluster 1:** Moderate life expectancy and healthcare investment.
3. **Cluster 2:** High-income countries with high life expectancy and health investment.

#### Recommendations

- **Cluster 0:** Prioritize international aid for healthcare.
- **Cluster 1:** Continue healthcare and infrastructure investments.
- **Cluster 2:** Maintain policies supporting health and economic stability.

### 2.6 Conclusion

This report offers insights into global socio-economic and healthcare disparities. By clustering countries based on indicators like income, child mortality, life expectancy, and GDP, we gain a clear understanding of differences that can guide policies, aid, and investment to support countries at varying development stages.

## Task 3: NBA Rookie Career Prediction Report

### 3.1 Introduction

The objective of this report is to predict which NBA rookies are likely to have careers lasting at least five years, a valuable insight for talent scouting and decision-making. Using the nba dataset, we explored three models: Logistic Regression, Gaussian Naive Bayes, and Neural Networks, each with unique strengths and limitations. Our target variable, Target\_5Yrs, identifies players with careers of 5+ years (1) or less than 5 years (0).

### 3.2 Understanding the Data and Selecting Features

The dataset includes performance metrics from rookies' seasons, such as Games Played, Minutes Played, and Field Goals Made. To identify the most relevant features, we examined correlations with the target variable. *Games Played*, *Minutes Played*, and *Points Per Game* were the top indicators of career longevity, forming the basis of our models.

### 3.3 Building and Evaluating the Models

The metrics used to evaluate the model are **Accuracy** (with values closer to 1 indicating higher accuracy) and **Number of misclassified points** (out of 447 total data points)

The codes for the models discussed here are available in the accompanying **Task 3\_Logistic**, **Task 3\_GNB** and **Task 3\_Neural Networks** python files respectively.

#### Model 1: Logistic Regression

- **Why This Model?** Logistic Regression is a straightforward, interpretable model often used for binary classifications (0 or 1) like ours.
- **How It Works:** Logistic regression assesses the probability that a player will have a long or short career, aiming to separate the data into these two categories.
- **Implementation: Features analyzed and Results:**
  - **Games Played:** Accuracy of 0.67; 149 misclassifications
  - **Games Played, Minutes Played:** Accuracy of 0.67; 146 misclassifications
  - **Games Played, Points Per Game, Rebounds:** Accuracy of 0.68; 142 misclassifications
  - The model suggests that rookies who play more games and score higher are more likely to have longer NBA careers.
- **Analysis:** Logistic regression provided a clear baseline, showing that features like Games Played and Points Per Game are meaningful. However, its linear assumption limits its ability to capture complex relationships in the data.

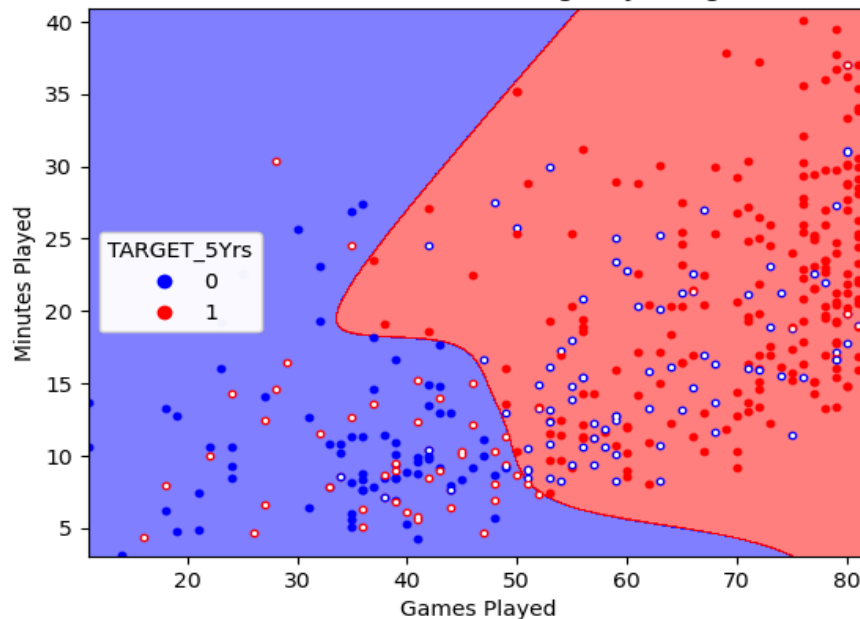
#### Model 2: Gaussian Naive Bayes

- **Why This Model?** Gaussian Naive Bayes offers a simple approach when we need fast predictions, especially with many features.
- **How It Works:** This model assumes each feature independently contributes to career predictions.
- **Implementation: Features analyzed and Results:**
  - **Field Goals Made:** Accuracy of 0.65, predicting a long career for players with more field goals
  - **Games Played, Minutes Played:** Accuracy of 0.65, predicting a short career for players with limited playtime
  - **Games Played, Minutes Played, Points Per Game, Field Goals Made:** Accuracy of 0.62
  - The model indicates that players with more field goals and game time are more likely to have longer NBA careers.
- **Analysis:** Naive Bayes was efficient but limited by its assumption of feature independence. It performed similarly to logistic regression but struggles with the correlated nature of basketball metrics, leading to accuracy constraints.

### Model 3: Neural Network

- **Why This Model?** Neural Networks are effective for detecting complex patterns and non-linear relationships, making them ideal when simpler models are not optimal.
- **How It Works:** By layering features in hidden layers, Neural Networks capture subtle interactions between features.
- **Implementation & Results:**
  - Games Played: Accuracy of 0.66, 151 misclassifications out of 447 points
  - Games Played, Minutes Played: Accuracy of 0.68, 143 misclassifications
  - Multiple high-correlation features: Accuracy of 0.67, 147 misclassifications
  - **Best Model:** Games Played and Minutes Played, 0.69 accuracy (with repeated random states for consistency)
  - The model performed best with Games Played and Minutes Played as features, achieving 0.69 accuracy and highlighting the importance of consistent playtime in predicting career longevity.

Neural Network Prediction of NBA Rookie Career Longevity Using Games and Minutes Played



**Figure 3:** Neural Network Prediction of NBA Rookie Career by Games and Minutes Played

Figure 3 above shows predictions based on Games Played and Minutes Played. Red and blue areas show the model's classifications for 5+ year and <5-year careers, respectively. Data points represent rookies, with red and blue circles indicating actual career outcomes. White stars mark misclassified points where predictions differ from reality.

### 3.4 Comparison and Recommendations

The Neural Network model performed best, though by a small margin, capturing subtle relationships that simpler models missed, making it the most reliable for career longevity predictions. Logistic Regression and Naive Bayes serve as useful baseline checks, but the Neural Network offers superior accuracy for long-term decisions.

In summary, the Neural Network provides a robust foundation for predicting rookie career longevity, making it the recommended model. Further tuning and feature exploration could enhance its accuracy.

## **Task 4: The Trolley Problem and Ethical Decision-Making in Autonomous Vehicles**

The Trolley Problem, a well-known ethical conundrum, explores the difficult choices involved in moral decision-making. In its simplest form, a runaway trolley is headed toward five people. An onlooker can choose to pull a lever, diverting the trolley onto a different track and saving five lives, but in doing so, sacrificing one person on the alternate track. This hypothetical scenario urges us to consider the moral implications of sacrificing one life to save others.

This moral dilemma gains practical relevance when applied to Autonomous Vehicles (AVs). Unlike human drivers, AVs are designed to make split-second decisions based on programmed algorithms rather than human intuition. Consider, for instance, a situation in which an autonomous vehicle must choose between endangering its passengers and swerving to avoid a pedestrian. This scenario highlights the challenge of programming AI to make moral decisions in real time (Millar, 2017).

One approach to this dilemma is to program AVs to minimize overall harm. While fewer lives may be lost, this approach can seem impersonal, particularly when it impacts innocent bystanders. Critics argue that ethics should consider concepts of justice and fairness, not just outcomes. Another perspective might be to restrict AVs from taking actions that directly harm innocent people, even if this increases the overall risk. This approach upholds the idea that certain actions are inherently wrong, regardless of their potential benefits.

To address these issues, researchers propose a data-driven strategy that considers public moral opinions. By collecting information on societal preferences across various scenarios, developers can create AVs that align more closely with social norms. Awad et al. (2018) found that people generally favor choices that protect the vulnerable and minimize harm. Integrating these societal values into AV programming could foster greater public trust.

However, the ethical challenges faced by AVs extend beyond the logic of the Trolley Problem. Issues of bias, liability, transparency, and data privacy also require attention. As AVs become more prevalent, it is essential for developers to engage in open dialogue with regulators, ethicists, and the public. This ongoing communication can build trust and help ensure that AVs are designed in a way that aligns with ethical and societal principles.

In summary, while the Trolley Problem provides a useful framework for exploring AV ethics, the complexities of real-world scenarios require a more nuanced approach. AV developers must balance safety, equity, and ethical considerations, while prioritizing public trust. As AI technology continues to evolve, ensuring that AVs adhere to societal norms will be crucial in creating a responsible and trustworthy automated future.

## 5. REFERENCES

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2. **Millar, J.** (2017) 'Ethics settings for autonomous vehicles', in **Lin, P., Keith, A., and Jenkins, R.** (eds) *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence*. New York: Oxford University Press, pp. 20–34.